

CHINSTRAP OPEN-SOURCE, MODULAR, ACCESSIBLE BOREHOLE INSTRUMENTATION PLATFORM

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INTRO

- Ice cores are the gold standard for paleoclimate study in the polar regions, but they're difficult to obtain. Projects are slow, risky, high carbon, and logistically complicated.
- We should leverage simpler rapid access ice drills for in situ data collection to enable smaller projects that target specific data points and get extra data from existing ice coring projects.
- Existing ice borehole loggers are not readily adaptable to new requirements and are bespoke to specific drilling systems. Developing new loggers is difficult, costly, and perceived as risky by funding bodies.

PROPOSAL

WE WANT TO CREATE AN OPEN-SOURCE MODULAR PLATFORM FOR ICE BOREHOLE INSTRUMENTATION

THE PLATFORM MUST BE SCIENTIST-FRIENDLY

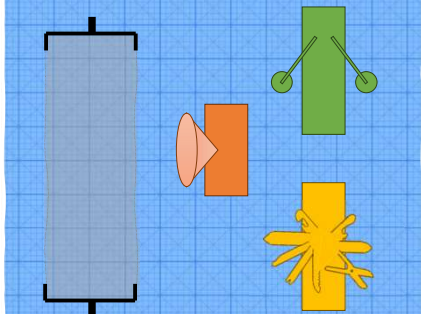
- Simple to develop new modules
- Simple to configure for specific projects
- Simple to control and analyse in the field

AND BENEFIT THE SCIENCE AND ENGINEERING COMMUNITIES

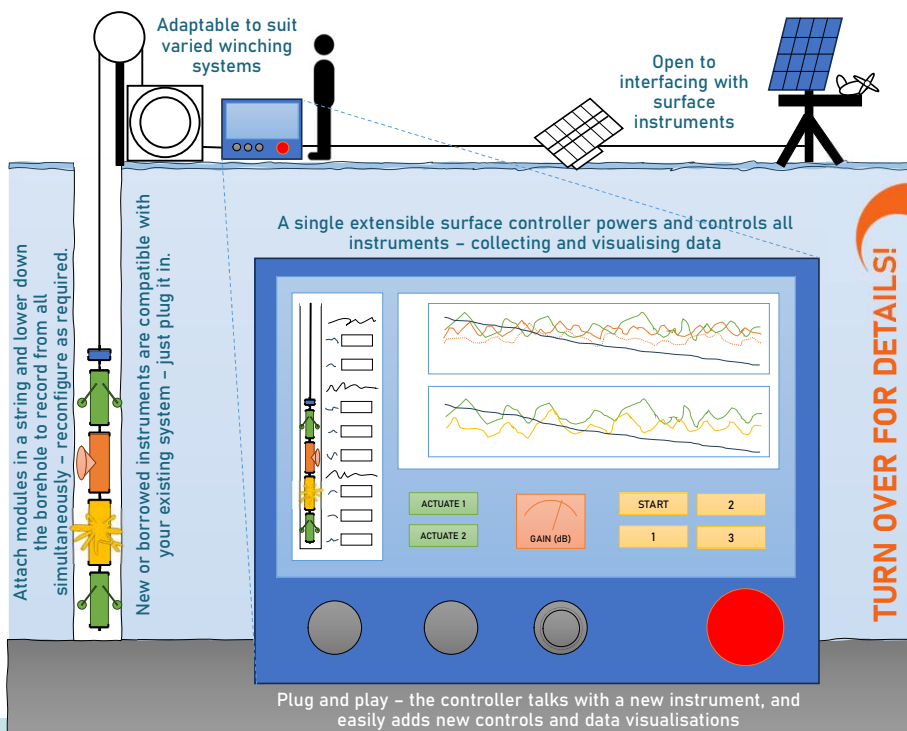
- Lower the barrier to entry for new science
- Share and copy instruments easily
- Do ground-breaking science with a smaller footprint

CREATE AN OPEN STANDARD TO SET:

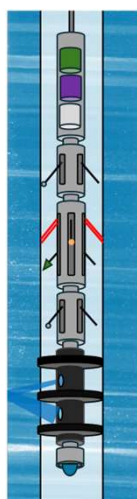
- The overall architecture, including power & comms
- Interfaces (mechanical, electrical, command & data)
- Instrument module design (a versatile chassis, standard PCBs, software libraries)



Meeting the platform standards when designing new instruments (or retrofitting existing sensors) allows reusing components. Ensuring designs are: field-ready, mechanically and electrically compatible, and integratable with minimal setup.



CURRENT FUNDING



1. Create the CHINSTRAP platform
2. Develop diameter, geometry, T, P, and AHRS module
3. Develop instrument centring module
4. Develop borehole Electrical Conductivity Measurement module
5. Develop optical scattering and vision module
6. Field test modules and platform
7. Open-Source: share, replicate, and expand

ICECOLD-CHRONO *

Oct 2026
- Nov 2028



SCIENCE GOALS

- Develop an instrument for rapid ice sheet chronology assessment in dry boreholes
- Borehole ECM to infer e.g. Annual Cycles, Volcanic Eruptions, Aerosol Flux
- Optical measurements to detect e.g. Annual Layers, Melt Layers, Dust
- Physical measurements to inform interpretation of above

TEST OUR LOGGER
AT YOUR FIELD
SITE!

* Ice Core Electrical Conductivity and Optical Logger for Downhole CHRONology

CHAT WITH US!

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- Which modules / instruments / measurements would interest you most?
- Where are the largest uncertainties in your models, and what downhole measurement(s) could best constrain them?
- Do you want short-term or long-term observation?
- What would put you off using the CHINSTRAP platform for your logger?



British
Antarctic Survey
NATURAL ENVIRONMENT RESEARCH COUNCIL

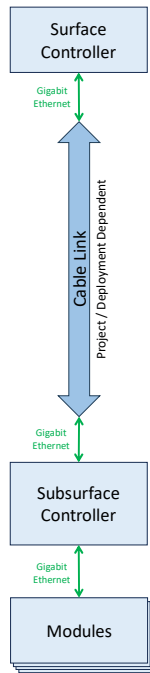
CHINSTRAP

HIGH LEVEL IMPLEMENTATION DETAILS

COMMS ARCHITECTURE

CHINSTRAP MUST WORK FOR ALL TEAMS

- **Compatible** with existing drilling systems
- **Adaptable and interchangeable** for different setups and cables
- **Future-proof** for new logger requirements and new drilling systems



PHYSICAL & DATA LINK ARCHITECTURE

Short runs – 1000BASE-T (Gigabit Ethernet)

Well supported, flexible and future-proof

- Allows **multiple parallel comms protocols** (e.g. video + control)
- Network access for **data sharing, remote access**, etc and to PCs for **development and debugging**
- Suitable electronics and connectors are **readily available** and **widely used** for subsea robotics, moorings, etc

Borehole cable – Blackboxed

Architecture flexible depending on cable used

- **Backwards compatible** with international community's drill cables
- Adaptors ensure **transparent link between controllers**
- Surface and Subsurface Controller command the modules whether there is a **'fast', 'slow', or no connection** to the surface.
- **Rate of data reporting controlled** to avoid overwhelming the borehole cable – ensuring high priority command and data gets through

		'No'	'Slow'	'Fast'
Physical Layer	No Connection			
		Power only	Twisted Pair	Optical Fibre
Data Link Layer e.g.	Preconfig / Autonomy	Powerline Modems	10BASE-T/IL CAN / RS485 VDSL2	Ethernet over Fibre
Application Layers		CBOR	→ ? → MQTT ? → RTSP / WebRTC	

APPLICATION LAYER (CONTROL & MONITORING)

Command & Simple Data – MQTT

- Lightweight **publish-subscribe messaging protocol** designed for constrained devices and **low-bandwidth networks**
- **Widely used** in hobby electronics so **accessible to amateur developers**
- **Robust** decoupled architecture

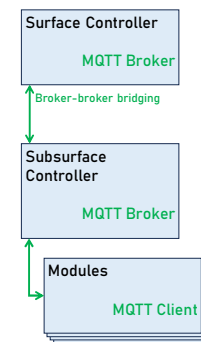
Surface and Subsurface Controllers are MQTT Brokers, Modules are MQTT Clients

Broker-broker bridging allows subsurface controller to take charge if link to surface fails or missing

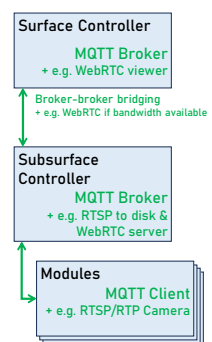
Additional Parallel Data Streams

Where bandwidth is available, **dense data streams** can be reported live to the surface using **protocols appropriate to the data type**. (e.g. live camera feeds, other continuous data)

Cmd and Simple Data over MQTT



+ parallel data streams e.g. camera



MODULE MECHANICS



MODULE CONNECTION

- **Rigid connection**
- **6-fold symmetry** provides orientation options
- **3-part design** allows removal for access without losing connection
- Choose between **isolated or continuous pressure vessels**



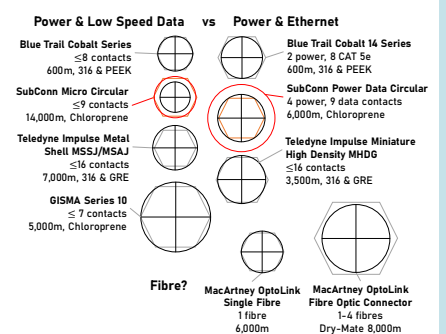
MODULE HOUSING

- Defined by cap geometry
- **Radial mounting** allows various module designs to attach
- **Length ambivalent**
- Face mounting points for interior components
- **Optional sealed environment** by O-ring seals and simple tubular pressure housing
- **Challenging to combine physical access to borehole with pressure seal**

ELECTRICAL CONNECTORS

REQUIREMENTS

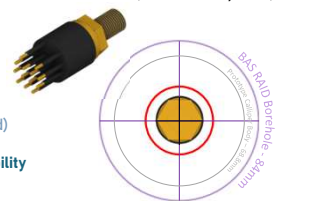
- **Small enough to fit** in smallest boreholes
- **Affordable and available**
- **Voltage and Current** rated
- **Enough conductors** for comms
- **Temperature and depth** ratings
- Versions available with drill fluid material compatibility



BEST FIT

SubConn Power Data Circular

- **35.5 mm shell** (1/2"-20 UNF thread)
- **Reasonable price** and lead time
- **600V, 8A**
- **13 pin** (4 data, 8 Gigabit Ethernet, shield)
- **-40°C, 6000m depth**
- **Material options** for drill fluid compatibility (increases cost and lead time)



STANDARDISE ON THREAD RATHER THAN THE CONNECTOR

- **1/2"-20 UNF** (with SubConn Power Data Circular recommended)
- **Ensures flexibility** for differing requirements (**Pressure Balanced Oil Filled connectors** or **subsea penetrators**)
- **Integrates best** with harshest environment (deep drilling fluid)
- **Use adaptors** for cheaper connectors for development, etc.